

Addition of CO by bubbling of CO through the solution using syringe

Date: 2022-12-07
Category: Protocol
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Goal

Bubble CO through a solution to enable the reaction of the CO with the substances in the solution

Prerequisites and preparation

Reaction flask

50 mL-Screw cap, threeway (120°) stopcock Schlenk flask

metal cannula

thin PVC-tube equipped with 1mL-Syringe

Transition piece: NS19 to tubing with stopcock

good septum

T-piece equipped with a threeway stopcock and two PVC-tubings

bubbler

Steps

Step number	Step description
1	Prepare the CO-flask: Therefore equip the 50 mL Screwcap Schlenk flask with a Screwcap and a new septum. Use the flask with the 3way stopcock, where the three openings are 120° tilted from each other and the fitting stopcock (two openings 120° from each other. Add a stirrer bar and set the flask under argon according to [Protocol] Schlenk Technique and heating the flask under vacuum three times.
2	Place the thin PVC-tube, which should be equipped with a 1 mL-syringe on the thinner connection of this flask. Equip it with a metal cannula and a stopper and set it under argon according to [Protocol] Schlenk Technique . See: CO flask.jpg
3	In argon counterstream the reaction flask is equipped with a transitionpiece NS19 to tubing. This is best done with the dried transitionpiece of used on the reflux condenser, or if no condenser is used the reaction flask is already prepared with the transitionpiece.
4	A septum (NS19) is set on the transitionpiece in argon counterstream. Subsequently the stopcock in the transitionpiece can be closed
5	The T-piece equipped with a threeway stopcock and two PVC-tubings is connected to the reaction flask. The open ending is connected to the Schlenk line and the part from the line to the flask is set under argon according to [Protocol] Schlenk Technique . See: Reaction flask.jpg

6	To the left over PVC tube a bubbler is connected. The bubbler should be placed as high as possible. To the bubbler another tube is connected to release the CO gas as high and close to the back into the fumehood. See: Setup bubbler 1.jpg and Setup bubbler 2.jpg The tubing from the line to the bubbler is flushed with argon for at least 30 s.
7	Both flask are placed so, that both can be stirred well (at least 300 rpm should be reached to assure a sufficient mixing)
8	In the CO flask the toluene (dry), formic acid (not dry) and acetic anhydride (not dry) are added. For this syringes are used and the screwcap is removed for each addition and immediately closed after each addition.
9	The metal cannula is put in the septum of the reaction flask and the volume between the septum and the stopcock of the transitionpiece is set under argon according to [Protocol] Schlenk Technique
10	The stopcock of the transitionpiece is opened and the cannula is placed in the solution.
11	The reaction flask is closed against the argon line using the T-piece and opened to the bubbler
12	To assure, that the setup works argon can be bubbled through the solution for a few seconds.
13	The CO flask is connected to the reaction flask. See Reaction setup.jpg
14	NEt ₃ (not dry) is added through the septum into the CO-flask. NEt ₃ should be added slowly to avoid any fume entering the reaction vessel. After the addition of approx. 0.2 to 0.7 mL the formation of bubbles should be observed in the CO flask and shortly after in the reaction flask and bubbler.
15	After the addition of all NEt ₃ it is waited a few minutes, till only small amounts of CO form.
16	To keep the reaction under CO the cannula is removed from the reaction flask and the stopcock of the transitionpiece is closed, as well as the stopcock directly on the reaction flask
17	The CO flask and the T-piece can be disassembled and cleaned.
18	To remove the CO the mixture is either directly set under vacuum, when the volume should be reduced or the reaction flask is flushed with argon for at least 30 sec.

Linked resource

Protocol - [Schlenk Technique](#)

Attached files

Reaction-setup.jpg

sha256: 96298a0452e96946ab455b97ab6148b2106c5a37a98b0072cbba381799f87b38



CO-flask.jpg

sha256: f42a7c3821c379301a5734a4d335296e5545a06e32b214f113a3d605d06d3360



Setup-bubbler-2.jpg

sha256: eb87d702bf3edd54547bd9834c2732e57a5f3d1d97216d179163cd26346f69c2



Setup-bubbler-1.jpg

sha256: 7f59a9f4eac90deb33d7e43341ea105ddc5df8d2d2bf0447ec63e4b97b2c3a56



Reaction-flask.jpg

sha256: d97c8b8d5c27955755247f8cce0f5f1936076bb912650f3375276c6ce012e47b



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